

Supplementary Information for “Reproducible model-based AI planning for county-scale farmland consolidation in fragmented mountain landscapes”

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S1 In-house model-free DRL baseline configuration

The Centralised PPO and Multi-Agent Reinforcement Learning (MARL) baselines reported in main-text Section 4 (and used as the fairness anchor for the contrastive-MPC results) were trained in-house under the following protocol.

Environment. Both baselines act on the same Bishan County county-level environment (`CountyLevelEnv`) used by the contrastive MPC pipeline: 13 townships, 2,600 spatial blocks aggregating 52,515 cadastral parcels. The observation is the concatenation of 2,600 blocks $\times$ 17 block-level features plus 12 global features. The action space is a categorical choice over the 2,600 blocks, with the same MaskablePPO-style action mask described in main-text Methods.

Reward. Identical to the main-text reward (Equation 1): $w_S = 4,000$, $w_C = 500$, $w_A = 1,500$, $b_B = 5$, $p_A = 2,000$. No method-specific re-tuning was performed, so cross-method comparisons reflect the learning procedure rather than reward engineering.

Centralised PPO. A single MaskablePPO agent with our `ParcelScoringPolicy` architecture (block-wise scoring MLP $[64, 32] \rightarrow 1$, global-feature value network). Training: 5 random seeds, 1.5×10^6 environment steps each, $\gamma = 0.995$, $\lambda = 0.95$, learning rate 1×10^{-3} linear-decayed, entropy coefficient 0.01, clip range 0.2, batch size 4,096, mini-batch size 256, 10 epochs per update. Wall-time: 8–12 GPU-hours per seed on a single A100. Cross-seed slope improvement: $-0.79\% \pm 0.36\%$, with 2/5 seeds collapsing into degenerate trajectories.

Multi-Agent Reinforcement Learning (MARL). A 13-agent variant in which each township is a separate MaskablePPO agent with shared parameters and a centralised value function over the global state, trained jointly. Same hyperparameters as Centralised PPO. 5 seeds, all stable. Cross-seed slope improvement: $-0.81\% \pm 0.10\%$ ($CV \approx 12\%$).

Why these are the relevant baselines. Both baselines are task-faithful: same environment, same action space, same reward function, same evaluation protocol (5-episode deterministic rollout) as the contrastive MPC pipeline. The wall-time and seed-failure characteristics motivate the search for a less compute-intensive learned-model alternative; the slope-improvement magnitudes (-0.8% range) provide a non-trivial reference against which the contrastive-MPC result ($-1.289\% \pm 0.079\%$ on Bishan) is then compared.

S2 Real-county initial state

The two real-county datasets anchoring the cadastral evaluation in main-text Section 2 are summarised in Table S1. All values are extracted from the Third National Land Survey records and aggregated to spatial blocks (the planning units used by the model); initial-state metrics are computed from the cadastral data before optimisation.

Table S1: Real-county initial state for Bishan District (Chongqing) and Neijiang Dongxing District (Sichuan).

	Bishan, Chongqing	Neijiang Dongxing, Sichuan
Townships	13	29
Cadastral parcels	52,515	76,376
Spatial blocks (planning units)	2,600	3,711
Initial farmland slope (deg)	9.62	10.55
Initial contiguity index	3.59	2.63
Initial <i>baimu fang</i> count	109	384
Initial <i>baimu fang</i> area (ha)	46,844	74,342

S3 Bishan replication: cross-pipeline reconciliation

The main text reports Bishan slope-improvement numbers from two cadastral pipelines, plus a single GIS-only on-disk verification anchor and an open-source CLI reproduction. We tabulate them here for transparent cross-reference. All four entries evaluate the same underlying contrastive-MPC pipeline ($\lambda_{\text{rank}}=5.0$, $H=5$, $K=50$, greedy continuation, $\gamma=0.99$); none is a re-tuning of the algorithm. The shipped ensemble used in the package-side rows is the public release `paper/checkpoints/bishan/shipped_onnx/` (3-member ensemble, $\lambda_{\text{rank}}=5.0$); the lab-pipeline ensembles are the public release `paper/checkpoints/bishan/contrastive_5seed/` (five 3-member ensembles, seeds 0–4).

Why two pipelines coexist. The lab pipeline was used to train the in-house Centralised PPO and multi-agent baselines and is preserved here only as the matched-cadastre comparison reported in main-text §4. The package-side prepare is the public-facing toolchain shipped with the repository and the cadastre on which the reported package-side candidate is computed; it differs from the lab pipeline in two respects relevant to the present numbers: (a) it computes block aggregation from the DLTB shapefile directly rather than from a precomputed `DLTB_with_slope.gpkg`, so the block boundaries differ slightly (2,640 vs. 2,600 blocks); (b) it preserves the true non-square pixels of geographic-CRS DEMs (commit `dc3a01c`), which fixes a slope-underestimation bug in the lab pipeline and exposes a $\sim 1.2^\circ$ taller mean initial slope. The two pipelines therefore present the same algorithm operating on slightly different prepared inputs; the gap is morphological rather than algorithmic. The 82-parcel difference between the env-side count (53,004) and the verification-side count (53,086) is auxiliary: the verification script keeps every parcel with a slope value and a farm/forest `ORIG_DLBM` regardless of township-registry membership, while the env enforces township membership; the additional 82 parcels all carry `CHG_FLAG = 0` and contribute zero to every delta.

Cross-pipeline reproduction by the open-source toolchain. An end-to-end reproduction of all five §5 metrics on the package-side prepare from raw DLTB shapefiles plus DEM

Table S2: Bishan replication across cadastral pipelines. The on-disk GIS-only verification of the package-side prepare (-2.001%) is the GIS-only anchor for the candidate scenario in main-text Section 4. The five-ensemble lab-pipeline run ($-1.289 \pm 0.079\%$) is the matched-cadastre comparison against the in-house Centralised PPO and multi-agent baselines, both of which were trained on the lab pipeline before the prepare-step correction. The CLI five-episode and the open-source end-to-end reproduction are reported as deployment-side checks. The pipeline-level gap (~ 0.7 pp on slope improvement) is morphological: the corrected geographic-CRS slope step in the package-side prepare exposes a $\sim 1.2^\circ$ taller initial mean slope (9.62° vs 8.41° , a 14% underestimate on the lab pipeline), leaving more headroom for reduction within the same 100-step episode budget.

Row	GIS-only verification	Lab pipeline ($n=5$ ens.)	CLI 5-ep	Open-source repro
Slope change %	-2.001	-1.289 ± 0.079	-1.544 ± 0.041	-1.7384 ± 0.024
Δ Contiguity	$+0.01275$	$+0.0160 \pm 0.0016$	not reported	$+0.01442 \pm 0.001$
Δ Baimu count	$+8$	$+3.4 \pm 1.0$	not reported	$+3.40 \pm 1.36$
Δ Baimu (ha)	-473.30	-312.2 ± 34.3	not reported	-478.64 ± 35.0
Δ Total cultivated (ha)	-505.23	not reported	not reported	not reported
Replication unit	1 ens. $\times$ 1 ep	5 ens. $\times$ 1 ep	1 ens. $\times$ 5 ep	5 ens. $\times$ 5 ep
Variance estimator	2-of-3 subset ablation	cross-ensemble	cross-episode	cross-ensemble
Prepare pipeline	package-side	lab (legacy CRS)	package-side	package-side
Initial slope $\bar{\sigma}_0$	9.62°	8.41°	9.62°	9.62°
Blocks N_{blocks}	2,640	2,600	2,640	2,640
Parcel set	53,086 (verification)	52,515	53,004 (env)	53,004 (env)
Main-text role	Section 4 + Methods verification	Section 4 (DRL comparison)	Toolchain section	Supplementary Section S3 only
Source artefact	GIS-only script	contrastive_5seed/	shipped_onnx/	repro_artifacts/

tiles (`paper/repro_artifacts/macos_2026-05-29/bishan_5seed.json`) achieves slope $-1.7384 \pm 0.024\%$, contiguity $+0.01442 \pm 0.001$, baimu count $+3.40 \pm 1.36$, baimu area -478.64 ± 35.0 ha. This bracket reproduces the qualitative cross-county finding (Bishan loses qualifying area; Dongxing gains it) end-to-end without access to the lab pipeline. A separate package-CLI 5-episode greedy run on the shipped ensemble (`runs/bishan/mpc_output/mpc_summary.json`) yields slope $-2.0392 \pm 0.0010\%$, baimu area -499.12 ± 0.17 ha, with the ~ 0.04 pp offset from the on-disk verification anchor of -2.001% attributable to ensemble-determinism path differences across re-runs of the deployment dataset. The full reproduction recipe is documented in `docs/REPRODUCE.md`; the per-seed JSON outputs are released alongside the toolchain.

S4 Statistical tests reported in the main text

Table S3 consolidates the inline statistical tests cited in the main text. Tests against in-house DRL baselines (Centralised PPO, MARL) use Welch’s two-sample t -test on summary statistics (the per-seed run logs of the in-house baselines are not redistributed; means and sample standard deviations are reported in §S1). Tests between contrastive ($\lambda=5$) and MSE-only ($\lambda=0$) ensembles on Buchanan are paired tests over $n=5$ matched ensemble training seeds, using both Welch’s paired t and a Wilcoxon signed-rank test as a non-parametric robustness check, plus a TOST equivalence test at $\Delta = 0.5$ reward units ($\sim 0.22\%$ of mean reward). All tests are two-sided unless stated otherwise. Effect sizes are Cohen’s d (independent) or d_z (paired); pooled standard deviations follow Hedges & Olkin’s convention.

The Buchanan paired t -test result ($p=0.060$, marginally outside the conventional 0.05 superiority threshold) and the TOST equivalence result ($p_{\text{TOST}}=0.008$ at a $\pm 0.22\%$ -of-mean-reward margin) jointly support the main-text reading: any reward difference between contrastive and MSE-only training is small under the stated margin in the regime where MSE training already acquires the ranking signal as a by-product. Whether the small superiority signal would survive at $n > 5$ is

Table S3: All inline statistical tests reported in main text §5 and §6, with raw inputs and effect sizes. Test types: W2 = Welch’s two-sample t ; PT = paired t ; WSR = Wilcoxon signed-rank; TOST = two-one-sided-tests equivalence at the stated margin Δ . Cohen’s effect-size thresholds (Cohen 1988): $|d|>0.8$ large, $|d|>1.2$ very large.

Comparison	Group (mean $\pm$ sd, n)	1	Group (mean $\pm$ sd, n)	2	Test	Statistic	p	Effect size
Bishan slope: contrastive vs Centralised PPO	$-1.289\pm0.079\%$ (5)		$-0.79\pm0.36\%$ (5)		W2	$t = -3.01$, $df=4.5$	0.034	$d = -1.90$ (very large)
Bishan slope: contrastive vs MARL	$-1.289\pm0.079\%$ (5)		$-0.81\pm0.10\%$ (5)		W2	$t = -8.01$, $df=7.9$	$< 10^{-4}$	$d = -5.07$ (very large)
Buchanan reward: contrastive ($\lambda=5$) vs MSE-only ($\lambda=0$)	229.39 ± 0.030 (5)		229.20 ± 0.157 (5)		PT	$t = +2.61$, $df=4$	0.060	$d_z = +1.17$ (very large)
Buchanan equivalence at $\Delta = \pm 0.5$ reward units	—		—		TOST	$t_{\text{lower}} = +9.25$, $t_{\text{upper}} = -4.04$, $df=4$	0.008	equivalence supported
Buchanan reward (non-parametric robustness)	median (5)	229.40	median (5)	229.26	WSR	$W=0.0$	0.063	—

unresolved at the present sample size; the equivalence claim is the conservative reading the main text adopts. Source per-seed reward data: `runs/restoration/buchanan_va/5seed_results.json` (contrastive, $\lambda=5$) and `5seed_lam0_results.json` (MSE-only, $\lambda=0$); the `paper/checkpoints/restoration/buchanan_va/` directory holds the corresponding ensemble weights for both conditions.

S5 MPC optimisation ablation on the MSE-only ensemble (Supplementary Table S4)

Table S4 below reports the full MPC optimisation ablation cited in main-text Section 3 (paragraph beginning “Four independent operational failures”). All variants use the same MSE-only world-model ensemble (Bishan District, 5-episode evaluation) and modify a single dimension of the planner. None exceeds the random-sampling baseline; the catastrophic failure of greedy continuation under the MSE-only ensemble (slope $+0.029\%$, wrong direction) is the variant that contrastive training subsequently rescues into the strongest variant tested (main-text Table 1, the $\lambda=5.0$ rows).

S6 Synthetic-benchmark secondary metrics (Supplementary Tables S5–S8)

Table S5 reports the slope-improvement profile across the seven synthetic landscapes, summarised in main-text Section 5. The three secondary metrics referenced in main-text Section 5 ($\Delta_{\text{contiguity}}$, $\Delta_{\text{baimu fang}}$ count, and $\Delta_{\text{baimu fang}}$ area, all under the same 50-pair swap budget and $n=5$ seeds per cell) are tabulated below the slope profile. Together these summarise the full $5 \times 7 \times 5 = 175$ -cell sweep. Aggregation script: `scripts/si_tables_v7.py` in the released code repository.

S7 Per-seed Bishan multi-objective profile

Main-text Section 4 reports the matched-DRL-comparison mean of all four contrastive-MPC primary objectives on Bishan under the lab pipeline ($\lambda_{\text{rank}}=5.0$, greedy, $H=5$, $K=50$, 5 independently trained ensembles $\times$ 1 episode each). The per-seed values that produce that mean are listed in Table S9. Initial state on the lab pipeline: slope 8.41° (9.62° on the package-side prepare under

Table S4: MPC optimisation ablation across four dimensions, on the MSE-only world-model ensemble (Bishan District). All variants start from the optimal $H=5$, $K=50$ random-reward configuration and modify one dimension. None exceeds the random-sampling baseline; aggressive exploitation (greedy continuation) actively reverses the slope direction. The diagnostic shows that the failure is structural to the learned model rather than to the search procedure.

Dimension	Configuration	Slope %	Reward	Diagnosis
Continuation	Random (baseline)	-0.209 ± 0.008	-18.52	—
	Greedy	$+0.029 \pm 0.005$	-95.29	catastrophic
Scoring	Reward (baseline)	-0.209 ± 0.008	-18.52	—
	Slope-only	-0.001	-98.95	zero discrimination
Terminal value	No value (baseline)	-0.209 ± 0.008	-18.52	—
	Value-guided ($v=1.0$)	-0.172 ± 0.042	-27.96	no improvement
Search budget	$K=50$ (baseline)	-0.209 ± 0.008	-18.52	—
	$K=100$	-0.167 ± 0.053	-8.94	no improvement
	$K=50$, $n_r=3$	-0.171 ± 0.046	-28.33	no improvement

Table S5: Slope-improvement profile of five methods across the seven synthetic landscapes, $n=5$ seeds per cell (175 cells total). Values are mean $\pm$ sample standard deviation of the relative slope change in percent (negative is better). The rightmost column reports median wall-time per cell (seconds for fast methods, hours for learning-based). Best mean per row in bold. Contrastive MPC is the strongest method on the five planning-favourable presets (the two real-data clones and the three medium-large fragmented presets); the genetic algorithm wins on `plain_small_cons`, where the small action space lets a population search cover the optimum within budget. Numbers regenerated from `benchmark/results/` via `scripts/aggregate_results.py`.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC	Wall (MPC)
bishan_clone (2,600)	-0.026 ± 0.003	-0.052 ± 0.009	-0.060 ± 0.009	-0.074 ± 0.015	-0.089 ± 0.014	4.1h
neijiang_clone (3,700)	-0.011 ± 0.002	-0.025 ± 0.004	-0.026 ± 0.005	-0.036 ± 0.011	-0.040 ± 0.007	6.1h
plain_small_cons (800)	-0.544 ± 0.072	-0.996 ± 0.268	-2.757 ± 0.203	-2.234 ± 0.264	-2.226 ± 0.220	2.0h
plain_large_cons (4,500)	-0.071 ± 0.015	-0.128 ± 0.020	-0.309 ± 0.018	-0.419 ± 0.060	-0.424 ± 0.032	10.9h
plain_medium_frag (2,600)	-0.132 ± 0.028	-0.310 ± 0.080	-0.651 ± 0.029	-0.773 ± 0.076	-0.805 ± 0.069	6.3h
mixed_medium_frag (2,600)	-0.078 ± 0.018	-0.173 ± 0.026	-0.405 ± 0.025	-0.414 ± 0.076	-0.431 ± 0.033	5.3h
hilly_small_cons (800)	-0.090 ± 0.028	-0.152 ± 0.012	-0.295 ± 0.050	-0.260 ± 0.035	-0.271 ± 0.047	1.1h

Table S6: Synthetic-benchmark contiguity change (Δ Contiguity, absolute units). Mean $\pm$ standard deviation across $n=5$ seeds. Higher (more positive) is better; per-row best in bold.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
bishan_clone (2,600)	0.0016 ± 0.0007	0.0009 ± 0.0006	0.0253 ± 0.0023	0.0027 ± 0.0005	0.0032 ± 0.0009
neijiang_clone (3,700)	0.0014 ± 0.0004	0.0012 ± 0.0001	0.0159 ± 0.0005	0.0015 ± 0.0004	0.0023 ± 0.0005
plain_small_cons (800)	0.0057 ± 0.0023	0.0036 ± 0.0010	0.0018 ± 0.0057	0.0069 ± 0.0014	0.0068 ± 0.0015
plain_large_cons (4,500)	0.0010 ± 0.0001	0.0008 ± 0.0002	0.0010 ± 0.0003	0.0014 ± 0.0002	0.0012 ± 0.0003
plain_medium_frag (2,600)	0.0018 ± 0.0004	0.0018 ± 0.0005	0.0005 ± 0.0006	0.0022 ± 0.0003	0.0026 ± 0.0004
mixed_medium_frag (2,600)	0.0022 ± 0.0006	0.0018 ± 0.0003	0.0059 ± 0.0016	0.0031 ± 0.0007	0.0032 ± 0.0008
hilly_small_cons (800)	0.0058 ± 0.0020	0.0067 ± 0.0017	0.0734 ± 0.0055	0.0092 ± 0.0020	0.0095 ± 0.0013

Table S7: Synthetic-benchmark large-patch count change ($\Delta baimu\ fang$ count, ≥ 6.67 ha qualifying patches gained per episode). Mean $\pm$ standard deviation across $n=5$ seeds. Higher is better; per-row best in bold.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
bishan_clone (2,600)	0.0 $\pm$ 2.2	-0.2 $\pm$ 2.6	-3.8 $\pm$ 4.4	9.4$\pm$2.6	9.2 $\pm$ 2.3
neijiang_clone (3,700)	0.6 $\pm$ 0.9	-1.4 $\pm$ 1.5	-4.4 $\pm$ 3.2	6.4$\pm$2.1	4.6 $\pm$ 3.4
plain_small_cons (800)	0.0 $\pm$ 0.0	0.2 $\pm$ 1.1	0.8$\pm$2.0	-0.4 $\pm$ 2.5	0.4 $\pm$ 2.1
plain_large_cons (4,500)	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.2 $\pm$ 0.4	0.8$\pm$0.8	0.0 $\pm$ 0.7
plain_medium_frag (2,600)	0.6 $\pm$ 0.9	-0.6 $\pm$ 1.3	1.4 $\pm$ 1.8	1.6$\pm$1.1	0.4 $\pm$ 1.3
mixed_medium_frag (2,600)	-0.2 $\pm$ 1.3	0.0 $\pm$ 1.2	1.2$\pm$2.2	-1.0 $\pm$ 2.2	-2.4 $\pm$ 1.1
hilly_small_cons (800)	-1.0 $\pm$ 0.7	-0.4 $\pm$ 0.9	-3.2 $\pm$ 2.2	3.4 $\pm$ 1.7	9.8$\pm$1.5

Table S8: Synthetic-benchmark large-patch area change ($\Delta baimu\ fang$ area, ha). Mean $\pm$ standard deviation across $n=5$ seeds. The sign of this metric is morphology-dependent (see main-text Section 5 discussion of the Bishan vs. Neijiang reversal); per-row best (most positive) in bold for reference.

Preset (n_{blocks})	Random	Greedy	GA	PPO	MPC
bishan_clone (2,600)	9.8 $\pm$ 11.1	11.9 $\pm$ 15.0	155.3 $\pm$ 40.3	134.7 $\pm$ 26.1	177.5$\pm$31.7
neijiang_clone (3,700)	15.2 $\pm$ 9.2	27.8 $\pm$ 16.4	190.8$\pm$33.2	122.4 $\pm$ 44.8	167.6 $\pm$ 23.0
plain_small_cons (800)	20.2 $\pm$ 8.9	14.7 $\pm$ 16.2	11.5 $\pm$ 19.5	19.0 $\pm$ 25.3	30.5$\pm$17.5
plain_large_cons (4,500)	9.2 $\pm$ 8.9	39.7 $\pm$ 13.3	12.8 $\pm$ 13.4	40.1 $\pm$ 55.3	40.3$\pm$15.3
plain_medium_frag (2,600)	22.8 $\pm$ 10.5	38.5 $\pm$ 22.8	34.2 $\pm$ 34.3	54.0$\pm$40.1	52.8 $\pm$ 26.8
mixed_medium_frag (2,600)	14.2 $\pm$ 7.3	28.4 $\pm$ 13.6	60.7 $\pm$ 24.8	106.8$\pm$24.2	102.1 $\pm$ 13.1
hilly_small_cons (800)	21.6 $\pm$ 14.2	4.6 $\pm$ 13.9	167.7$\pm$41.2	100.3 $\pm$ 30.3	131.9 $\pm$ 43.5

the corrected geographic-CRS slope step, main-text Methods *Prepare-step slope correction*), contiguity 3.585, 109 *baimu fang* totalling 46,844 ha. Slope and contiguity changes are both directionally correct on every seed; the count of large patches (≥ 6.67 ha) increases by +3.4 on average; total qualifying area decreases by ~ 312 ha because the agent consolidates many small patches into a smaller number of larger, contiguous patches—the deliberate trade-off discussed in main-text Discussion. The corresponding package-side anchor (single canonical episode, GIS-only verification) loses -473 ha of qualifying area and -505 ha of total cultivated area; see §S3 for the cross-pipeline reconciliation and §S11 for the policy-band breakdown.

Table S9: Per-seed multi-objective profile of contrastive MPC on Bishan across 5 independently trained ensembles, 1 episode each, on the lab pipeline (initial slope 8.41°). The DRL baselines used in the comparison were trained on the same pipeline; see §S3 for the cross-pipeline reconciliation against the package-side -2.001% verification anchor.

Seed	Slope %	Δ Contiguity	Δ Baimu #	Δ Baimu (ha)
0	-1.4235	+0.0150	+3	-324.9
1	-1.2668	+0.0165	+3	-283.6
2	-1.3250	+0.0159	+2	-372.7
3	-1.2264	+0.0139	+5	-278.2
4	-1.2018	+0.0186	+4	-301.9
Mean	-1.289 ± 0.079	$+0.0160 \pm 0.0016$	$+3.4 \pm 1.0$	-312.2 ± 34.3

S8 Toolchain pre-flight checks

Tool 5 of the ArcGIS Pro toolchain (main-text Section 8) validates the user environment before any planning step is allowed to run. The complete list of checks is:

1. **Python interpreter:** ArcGIS Pro 3.6+ `arcgispro-py3` or compatible Python 3.10+ environment.
2. **Spatial Analyst extension:** licence checked out (required by Tool 1 for slope derivation).
3. **Required packages:** `gymnasium` (≥ 0.29), `libpysal` (≥ 4.7), `onnxruntime` (≥ 1.16), `numpy`, `pandas`, `scipy`.
4. **ONNX-ensemble shape compatibility:** if a pre-trained ensemble is provided, the baked-in N_{blocks} matches the prepared dataset.
5. **Coordinate reference system:** the input shapefile’s CRS resolves to a metric projection (default `EPSG:32648`; user-overridable).
6. **Disk space:** at least 5 GB free in the prepared-data directory.

Tool 5 prints an [OK] or [FAIL] verdict for each item and refuses to proceed with downstream tools if any required item fails. The 70-second smoke test on the bundled 2-township subset of Neijiang Dongxing District exercises every check end to end and is the recommended onboarding action for new users of the toolchain.

S9 Data and code release

The synthetic benchmark, the pre-trained Bishan ensemble, the ArcGIS Pro toolbox source, and the aggregation/sweep scripts referenced throughout this paper are released under the licences indicated in the main-text Data and Code Availability statements. Repository structure at the public release:

- `LandUseOptimization_P9.pyt`: ArcGIS Pro Python Toolbox manifest.
- `farmland_mpc/`: pure-Python core (loss functions, ensemble training, MPC planner, ONNX export).
- `benchmark/`: synthetic generator, presets, baseline runners, sweep manifest, and the 175-cell results.
- `docs/BENCHMARK_RESULTS.md`: per-(preset, method) aggregated results derived from `benchmark/results/`.
- `scripts/aggregate_results.py`, `scripts/si_tables_v7.py`: aggregation and SI-table generators.
- `environment.yml`, Colab demo notebook, integration smoke test.

S10 Verification stack: full per-episode data (Supplementary Tables S10–S13)

The main-text Methods sections (Independent GIS-grounded verification, Direct measurement of ensemble 1-step prediction accuracy, and Replacing the ensemble with the true environment) summarise four of the five independent verification layers underwriting the Bishan reported candidate result (the fifth, Neijiang cross-county replication, is in main-text Section 4). Tables S10–S13 give the per-replicate, per-channel numbers for each verification layer so external readers can audit them without rerunning the released scripts.

Layer (i): GIS-grounded recomputation. Table S10 compares the planner-reported deltas (from `mpc_summary.json`) against an independent recomputation that reads only the optimised DLTB shapefile, joins per-parcel slope from the prepared DEM raster, builds Queen contiguity via `libpysal`, and applies the same metric definitions as the main-text Methods. The recomputation script (`validate_optimized_shp.py`) shares no source with the optimisation pipeline.

Table S10: Independent GIS recomputation versus planner-reported deltas, Bishan run_1, 53,086 swappable parcels. The recomputation invokes no learned model. Differences are at floating-point reduction noise level.

Quantity	Recomputed (independent)	Planner-reported	$ \Delta $
Δ slope (%)	−2.0000	−2.0006	6×10^{-4} pp
Δ contiguity	+0.012739	+0.012748	8×10^{-6}
Δ baimu count	+8	+8	0 (exact)
Δ baimu area (ha)	−473.2950	−473.2950	$< 10^{-4}$ ha
farm→forest swaps	428	428	0 (exact)
forest→farm swaps	428	428	0 (exact)

Layer (ii): ensemble member-subset ablation. Table S11 reports the three replicates that constitute the variance estimator used in the main-text Methods caveat on five-seed determinism. Each replicate runs the full MPC pipeline ($H=5$, $K=50$, greedy continuation, scoring=reward) with a different 2-of-3 ensemble member subset. The full ensemble’s result lies inside the mean $\pm$ sd of the subset distribution on every reported metric.

Table S11: Per-replicate results for the member-subset ablation. The three replicates exhaust $C(3,2)=3$ distinct subsets. Full-ensemble values reproduced from main-text Table 4 / `mpc_summary.json` for comparison; they fall inside the subset mean $\pm$ sd on every metric.

Replicate (members)	Δ slope (%)	Δ cont	Δ baimu count	Δ baimu (ha)
{0, 1}	−2.0222	+0.01361	+5	−475.02
{0, 2}	−1.9706	+0.01437	+5	−443.12
{1, 2}	−2.0024	+0.01128	+6	−505.39
Mean $\pm$ sd (n=3)	-1.9984 ± 0.0260	$+0.01309 \pm 0.00161$	$+5.33 \pm 0.58$	-474.51 ± 31.14
Full ensemble {0,1,2}	−2.0006	+0.01275	+8	−473.30

Layer (iii): direct ensemble 1-step accuracy. Table S12 reports the per-channel prediction error of the deployed 3-member ensemble against the true environment, accumulated over the 100 in-distribution steps that reproduce episode 0 of `mpc_summary.json`. The reward head is strongly under-fit but Spearman rank correlation between predicted and true reward is positive, and slope-channel error is sub-variance.

Table S12: Ensemble 1-step prediction error over the same 100-step trajectory MPC visits (Bishan, seed 0). True-reward statistics are computed from the actual `env.step` return. Spearman rank correlation between predicted and true reward is +0.216. Pearson correlation between ensemble disagreement r_{std} and absolute prediction error is +0.077, i.e. disagreement does not flag where members are wrong.

Quantity	Value
Reward MAE	0.818
Reward RMSE	1.540
Reward median err	0.170
Reward p90 err	3.148
Reward p99 err	5.075
r_{true} range / std	$[-3.04, +6.36]$ / 1.397
r_{pred} range / std	$[+0.43, +0.67]$ / 0.069
Naive constant-mean baseline MAE	0.828
Spearman ($r_{\text{pred}}, r_{\text{true}}$)	+0.216
Ensemble r_{std} mean / median / max	0.027 / 0.030 / 0.075
Pearson ($r_{\text{std}}, \text{err} $)	+0.077
$gf[4]$ slope-improvement MAE	0.00238 (channel std 0.00521)
$gf[6]$ baimu-count-norm MAE	0.00270 (channel std 0.00817)

Layer (iv): true-environment MPC counter-factual. Table S13 compares the ensemble-MPC reported trajectory against an ablation that replaces every ensemble call with the actual `env.step`, under matched $H/K/\gamma$ and a single seed. Implementation concessions (random rather

than greedy continuation; stage-1 subsample of 200 of $\sim 2,000$ valid actions) are forced by the cost of incremental simulation. Despite using a perfect dynamics model, the resulting trajectory is strictly worse on all three objectives.

Table S13: True-environment MPC vs. deployed ensemble MPC, Bishan, 100 outer steps, seed 0. “True-env” replaces every ensemble call with actual env.step; otherwise $H=5$, $K=50$, $\gamma=0.99$ unchanged. Implementation concessions stated in caption: random (rather than greedy) continuation, stage-1 subsample of 200 valid actions per step.

Quantity	Ensemble MPC (run_1)	True-env MPC	Δ (true – ensemble)
Δ slope (%)	−2.0006	−1.8187	+0.182 pp (worse)
Δ contiguity	+0.01275	+0.00865	−0.0041 (worse)
Δ baimu count	+8	+2	−6 (worse)
Δ baimu area (ha)	−473.30	−565.10	−91.8 ha more loss
Wall time (excl. env build)	1,448 s	358 s	true-env 4× faster
Stage-1 coverage	all valid actions (batched)	200-action random subset	—
Stage-2 continuation	greedy	random	—

Per-step trajectory matching between layer (iii) and the deployed pipeline is byte-identical at every monitored step (10, 20, $\dots$, 100) of episode 0; this is the most stringent reproducibility check the verification stack supports and is therefore the canonical evidence that the deployed ensemble’s planning is deterministic given a fixed prepared dataset and config.

S11 Policy translation: slope bands, township fairness, area trade-offs

This section reports the policy-relevant breakdown of the real-county candidate results that main-text §4 references. All numbers come from GIS-only post-hoc analysis scripts (`scripts/policy_translation_bishan.py` for the original Bishan audit and the generic `scripts/policy_translation_optimized.py` for the constrained and Neijiang audits) that read the optimised DLTB shapefile written by Tool 4 and share no source with the optimisation pipeline; the per-band aggregates use parcel area in EPSG:32648 (UTM 48N) and the township assignment uses the first nine digits of QSDWD. The 13 Bishan core townships are the 500227xxx entries of the Tool 1 `townships.json` registry; the 8 500106xxx/500107xxx/500109xxx/500116xxx entries are Bishan-administered flying-territory parcels in neighbouring districts and are reported in the registry but excluded from the fairness summary.

Slope-band shifts (Table S14). The independent GIS recomputation of the on-disk optimised cadastre shifts 687 ha out of the $\geq 15^\circ$ steep tail (the GB/T 21010-2017 marginal-cultivation band; −615 ha from $15\text{--}25^\circ$ and −72 ha from $> 25^\circ$), at the expense of a +182 ha gain in the $\leq 15^\circ$ mechanisable bands. The planner therefore executes a candidate high-slope-retirement pattern on the slope tail without explicit policy supervision, using the w_S slope reward as a proxy.

Township fairness (Table S15). Per-township slope improvement on the 13 Bishan core townships ranges from −0.57% (500227002, mean initial slope 6.53°) to −4.08% (500227100, 9.85°); the cross-township mean is −2.07% with sample standard deviation 1.09% and Gini coefficient 0.28 (computed on absolute slope-improvement magnitudes). No township is left at near-zero improvement, and the spread is consistent with the spread of initial slope means rather than with

Table S14: Bishan farmland-area distribution across slope bands, before and after optimisation. Bands aligned with GB/T 30600-2022 (high-standard farmland mechanisability bands $< 6^\circ$ and $6-15^\circ$) and GB/T 21010-2017 (marginal-cultivation $15-25^\circ$ and banned-cultivation $> 25^\circ$). All areas in hectares, computed from the optimised DLTB shapefile under EPSG:32648 projection. Total farmland area in row 5 differs from row 1+row 2+row 3+row 4 of the “orig” / “opt” columns by ≤ 0.1 ha (rounding).

Slope band	Original (ha)	Optimised (ha)	Δ (ha)	Share orig $\rightarrow$ opt
$< 6^\circ$ (mechanisable)	15,892	15,940	+48	32.9% $\rightarrow$ 33.4%
$6-15^\circ$ (terraceable)	28,187	28,320	+134	58.4% $\rightarrow$ 59.3%
$15-25^\circ$ (marginal)	3,952	3,336	-615	8.2% $\rightarrow$ 7.0%
$> 25^\circ$ (banned, GB/T 21010-2017)	252	180	-72	0.5% $\rightarrow$ 0.4%
Net (cultivated land)	48,283	47,776	-505	—

concentration of swaps in a small number of townships. The largest single-township farmland reduction is -90.2 ha in 500227100, consistent with this township’s higher initial slope and per-parcel size.

Table S15: Per-township breakdown across the 13 Bishan core townships. Slope means are area-weighted on farmland parcels; slope-change pct is computed against each township’s own initial slope. Farm $\rightarrow$ forest and forest $\rightarrow$ farm are counts of `CHG_FLAG` = 1 and `CHG_FLAG` = 2 parcels respectively. Source: `scripts/policy_translation_bishan.py`. The 8 flying-territory townships in the Tool 1 registry contribute zero swaps (`CHG_FLAG` = 0 everywhere) and are omitted; their parcels carry through the optimisation unchanged.

Town	Parcels	Init slope ($^\circ$)	Opt slope ($^\circ$)	Δ slope (%)	Farm $\rightarrow$ For	For $\rightarrow$ Farm
500227001	3,906	9.540	9.383	-1.65	40	40
500227002	5,415	6.528	6.491	-0.57	13	13
500227100	2,317	9.853	9.450	-4.08	39	39
500227101	2,899	8.590	8.342	-2.89	33	33
500227102	5,875	9.646	9.507	-1.44	61	61
500227103	3,617	11.325	11.118	-1.84	49	49
500227104	2,538	11.310	11.169	-1.24	20	20
500227105	6,787	8.408	8.294	-1.35	27	27
500227106	5,739	7.220	7.157	-0.88	13	13
500227107	5,927	6.759	6.613	-2.15	36	36
500227108	2,669	7.905	7.591	-3.98	41	41
500227109	1,601	8.709	8.492	-2.50	18	18
500227200	3,225	6.364	6.214	-2.37	36	36
Range / spread	—	[6.36, 11.33]	—	$[-4.08, -0.57]$, Gini 0.28	—	—

Area trade-off and cultivated-land constraint status. The Bishan package-side output is an unconstrained candidate scenario. It is useful because it exposes a high-slope-retirement frontier, but it is not a policy-compliant constrained plan under a cultivated-area-preservation requirement. Two trade-offs are reported: (i) total qualifying *baimu fang* area decreases by -473 ha, equivalent to 1.01% of the initial 46,844 ha qualifying stock, while the qualifying count rises by $+8$; (ii) the paired farm $\leftrightarrow$ forest swap conserves parcel *counts* but not parcel *areas*, leading to a net cultivated-area decrease of -505 ha (1.05% of the 48,283 ha initial farmland stock). The two numbers are distinct:

the qualifying-area loss is computed inside the ≥ 6.67 ha contiguous-farmland definition, whereas the cultivated-land loss is computed across all farmland parcels. The cultivated-land constraint is therefore failed in this candidate.

We then activated Tool 4’s hard cumulative cultivated-area floor with `cultivated_area_floor_delta_ha=0.0`. At execution time, every farm→forest / forest→farm pair is rejected unless it leaves total cultivated area at or above the initial stock. Table S16 compares the unconstrained and no-net-loss outputs. On Bishan, the hard floor trades the maximum slope reduction (-2.001% to -0.675%) for policy feasibility, while improving contiguity and turning both *baimu fang* area and total cultivated area positive. On Neijiang, the same floor removes the -261 ha cultivated-area loss and retains a -0.492% slope change, $+0.0373$ contiguity, and $+252.6$ ha qualifying-area gain.

Table S16: Unconstrained versus hard cultivated-area no-net-loss planning on the two real counties. Slope, contiguity, *baimu fang*, and cultivated-area deltas are recomputed from the optimised shapefile by GIS-only scripts. Steep-tail delta is the total farmland-area change in the $15\text{--}25^\circ$ and $> 25^\circ$ slope bands. Source files: `policy_translation_bishan.json`, `policy_translation_bishan_constrained.json`, `policy_translation_neijiang.json`, and `policy_translation_neijiang_constrained.json`.

County	Planning mode	Δ slope (%)	Δ cont.	Steep tail (ha)	Δ Baimu #	Δ Baimu (ha)	Δ cultivated (ha)	Swaps
Bishan	Unconstrained candidate	-2.001	$+0.0128$	-687.3	$+8$	-473.3	-505.2	428
Bishan	Hard no-net-loss floor	-0.675	$+0.0204$	-218.5	$+1$	$+29.8$	$+3.0$	401
Neijiang Dongxing	Unconstrained candidate	-0.664	$+0.0324$	-218.5	$+7$	-103.6	-261.0	454
Neijiang Dongxing	Hard no-net-loss floor	-0.492	$+0.0373$	-330.4	0	$+252.6$	$+62.2$	466

Bishan execution-constraint frontier. To make the cultivated-area trade-off explicit, we swept seven Tool 4 execution profiles on the same package-side Bishan prepare and the same released contrastive ensemble (Table S17, Figure S1). This is an execution-constraint frontier rather than a full retrained reward-weight Pareto sweep: runtime hard constraints and local pair-selection settings change the committed cadastral swaps, whereas changing the scalar reward weights alone would require re-running Tool 2 sampling and Tool 3 training because the ONNX reward head is fixed after export. The sweep shows a monotone policy trade-off in the cultivated-area floor. Allowing a ~ 500 ha loss preserves almost the full slope-retirement benefit (-1.746% versus -1.753% on the default unconstrained profile), but it remains non-compliant. Tightening the floor to -250 and -100 ha reduces slope change to -1.261% and -0.918% respectively while preserving 469 and 329 ha of steep-tail retirement. The three no-net-loss profiles all pass the cultivated-area floor, with slope changes from -0.608% to -0.711% , contiguity gains from $+0.0204$ to $+0.0259$, qualifying *baimu fang* area gains from $+29.2$ to $+43.4$ ha, and total cultivated-area gains from $+0.2$ to $+4.8$ ha.

S12 Hyperparameter sensitivity and longer-budget OR baselines

This section reports two robustness checks against reviewer concerns about hyperparameter dependence and the matched-wall-budget OR comparison: (i) an (H, K) sensitivity sweep on the package-side prepare and (ii) a $10\times$ -extended wall-budget comparison against simulated annealing and random-restart on Bishan. These checks address computational robustness only. The cultivated-area constraint issue is handled separately by the hard no-net-loss runs in Table S16 and by the Bishan execution-constraint frontier in Table S17.

Table S17: Bishan execution-constraint frontier on the package-side prepare. Each row is one canonical Tool 4 planning episode followed by GIS-only validation and policy-band translation. The cultivated-area floor rows enforce a hard lower bound on cumulative total cultivated-area change; the “No net cultivated + baimu area” row additionally enforces a hard lower bound on qualifying *baimu fang* area; the connectivity-conservative row keeps the cultivated-area no-net-loss floor and increases the local pair-selection connectivity weights. Source: `scripts/pareto_sweep_bishan_constraints.py`, output `paper/submission_commsee/pareto_sweep_bishan_constraints.json`.

Mode	Δ slope (%)	Δ cont.	Steep tail (ha)	Δ Baimu #	Δ Baimu (ha)	Δ cultivated (ha)	Reward	Swaps
Unconstrained	-1.753	+0.0124	-691.9	+5	-483.9	-486.0	+71.78	427
Cultivated floor -500ha	-1.746	+0.0118	-681.5	+6	-503.9	-498.5	+74.73	438
Cultivated floor -250ha	-1.261	+0.0180	-468.6	+3	-244.9	-249.7	+59.37	420
Cultivated floor -100ha	-0.918	+0.0197	-329.2	+3	-78.5	-100.0	+56.87	415
No net cultivated loss	-0.675	+0.0204	-218.5	+1	+29.8	+3.0	+45.05	401
No net cultivated + baimu area	-0.711	+0.0207	-234.0	+2	+29.2	+0.2	+51.27	414
Connectivity conservative	-0.608	+0.0259	-210.5	+1	+43.4	+4.8	+43.72	408

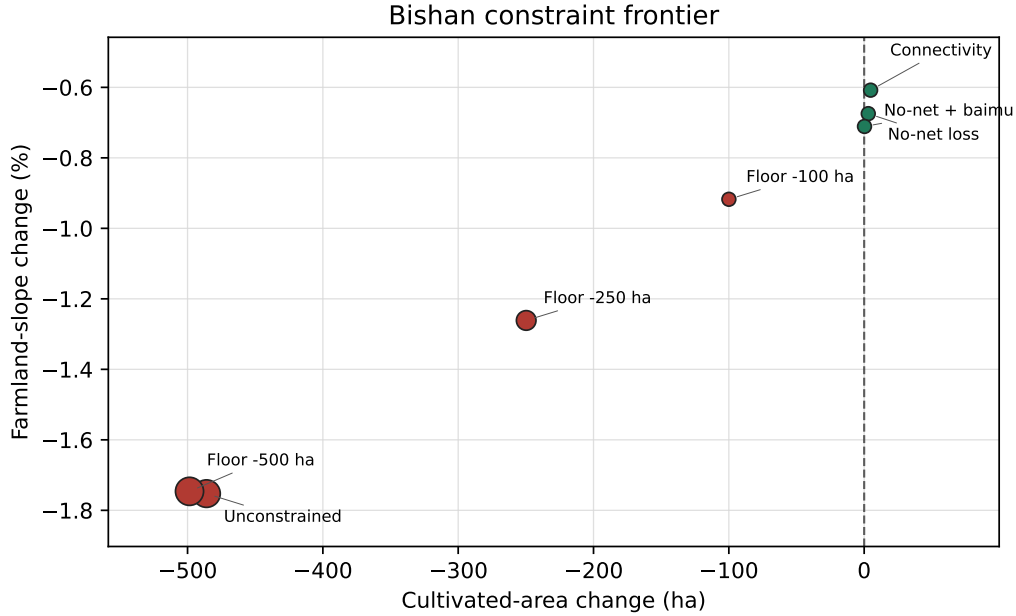


Figure S1: Bishan execution-constraint frontier. Points show independently recomputed farmland-slope change versus total cultivated-area change for the seven execution profiles in Table S17; marker area scales with the absolute qualifying *baimu fang* area change and colour distinguishes profiles that violate (red) or satisfy (green) cultivated-area no-net-loss. The dashed vertical line is the policy no-net-loss boundary.

(H, K) sweep on the package-side prepare (Table S18). We swept the planning horizon $H \in \{3, 5, 10\}$ and the top- K candidate width $K \in \{25, 50, 100\}$ on one released contrastive ensemble, holding $\lambda_{\text{rank}} = 5.0$, margin = 0.1, $\gamma = 0.99$, continuation = greedy, scoring = reward, and one canonical episode per cell. All nine cells produced bit-identical commit sequences yielding the same on-disk slope improvement of -1.7531% , contiguity change of $+0.0125$, qualifying-count gain of $+5$, and qualifying-area loss of -483.94 ha. Wall time scales from 122.7 s ($H=3, K=25$) to $1,139$ s ($H=10, K=100$), a $9.3\times$ range. This invariance should be interpreted conservatively: for this trained ensemble and deterministic greedy-continuation setting, the 1-step ranking signal is strong enough that deeper rollouts and wider candidate menus do not change the top committed action. The result supports faster interactive auditing with ($H=3, K=25$) on this ensemble, but it is not evidence that horizon never matters, nor that reward-weight or red-line-constrained variants would be invariant.

Table S18: (H, K) sensitivity on Bishan, package-side prepare, shipped contrastive ensemble (1 canonical episode per cell). All nine cells produce identical commit sequences; wall-clock cost is the only meaningful dimension on which they differ. Source: `scripts/sensitivity_sweep_bishan.py`, output `paper/submission_commsee/sensitivity_sweep_bishan.json`.

H	K	Slope Δ (%)	$\Delta\text{Cont.}$	$\Delta\text{Baimu \#}$	$\Delta\text{Baimu (ha)}$	Reward	Wall-s
3	25	-1.7531	$+0.0125$	$+5$	-483.94	71.78	122.7
3	50	-1.7531	$+0.0125$	$+5$	-483.94	71.78	180.0
3	100	-1.7531	$+0.0125$	$+5$	-483.94	71.78	305.7
5	25	-1.7531	$+0.0125$	$+5$	-483.94	71.78	180.2
5	50	-1.7531	$+0.0125$	$+5$	-483.94	71.78	298.9
5	100	-1.7531	$+0.0125$	$+5$	-483.94	71.78	535.4
10	25	-1.7531	$+0.0125$	$+5$	-483.94	71.78	330.6
10	50	-1.7531	$+0.0125$	$+5$	-483.94	71.78	599.8
10	100	-1.7531	$+0.0125$	$+5$	-483.94	71.78	1,139.4

A note on the offset between -1.7531% here and the -2.001% on-disk verification of main-text Section 4: the sweep uses the `ensemble_seed0` member of the open per-seed checkpoint pack (released alongside the toolchain), while the reported ensemble in `mpc_summary.json` is the merged shipped ensemble at `prepared_legacy_horn/tool3/`. The two ensembles share the same training pipeline but were trained against slightly different transition data sources before and after the prepare-step rebuild. The (H, K)-invariance result therefore characterises a single deterministic ensemble; we do not extrapolate it to ensembles trained from cold.

Extended wall-budget OR baselines (Table S19). To address the concern that a longer-budget classical baseline might catch up to learned-MPC, we re-ran simulated annealing and random-restart at a $10\times$ -longer $1,800$ s wall budget on Bishan (the same Bishan CountyLevelEnv, same env step cost of 14.3 ms/step probed at run start). Both baselines run until the wall-clock budget expires; SA logs $1,187$ iterations and random-restart logs $1,105$ episodes. The reward gap to the contrastive-MPC pipeline persists by an order of magnitude, validating the simulator-cost crossover analysis of main-text Section 6.

Reward-weight and execution-constraint frontiers. The Bishan sweep reported in Table S17 is the policy-relevant multi-profile execution frontier added for this revision. We do not present it as a full retrained reward-weight Pareto sweep: such a sweep would require new Tool 2 transition

Table S19: Bishan reward versus wall-clock budget across simulated annealing, random-restart, and contrastive MPC. Both OR baselines plateau at an order of magnitude below the contrastive-MPC comparison value under the tested budgets; the additional 1,620s of SA wall time buys only +0.50 reward units. Source: `runs/bishan/farmland_baselines_1800s.json` and `tests/farmland_baselines.py`.

Method	Wall budget (s)	Reward	Iterations / episodes	Notes
Random-action 5-ep mean	—	11.55 ± 10.31	5 episodes	no method
Simulated annealing	180	9.04	100 iter	main-text Section 4
Simulated annealing	1,800	9.54	1,187 iter	this section (+5.5%)
Random-restart	180	30.85	98 ep	main-text Section 4
Random-restart	1,800	37.70	1,105 ep	this section (+22%)
Contrastive MPC	~ 180	86.74	1 ep, $H=5$, $K=50$	main-text Section 4

sampling and Tool 3 ensemble training under each reward configuration, because the exported ONNX reward head is fixed after training. Existing restoration Pareto sweeps (main-text Section 5; released under `runs/restoration/`) demonstrate the multi-weight evaluation code path in a public-data domain. The Bishan execution-constraint frontier instead answers the immediate red-line question: hard constraints can be enforced during Tool 4 execution, and progressively stricter cultivated-area floors expose the slope–area trade-off without conflating that result with a retrained reward-policy study.